FALK TRUE TORQUE

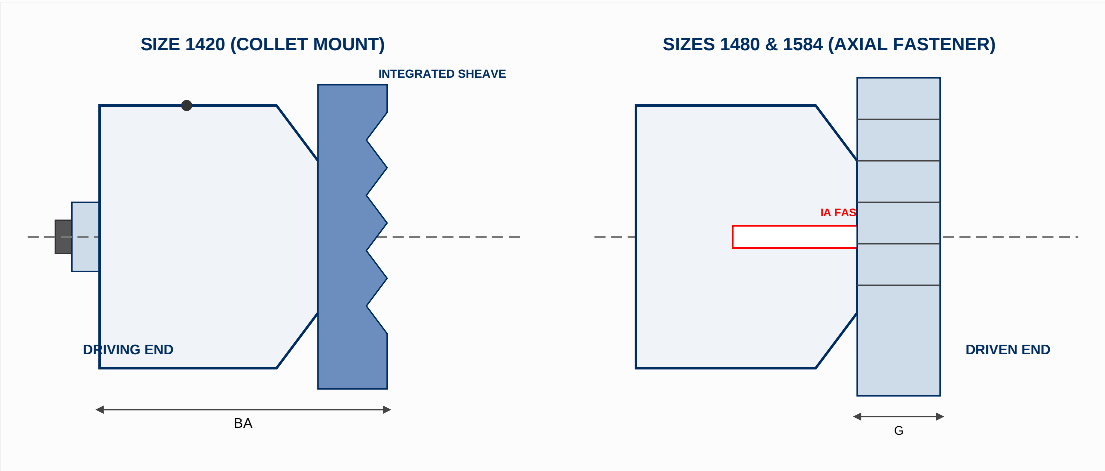
Fluid Couplings – Heavy Duty HF41 Sheave/Hollow Shaft Mount (Sizes 1420 – 1584)

The Falk True Torque Type HF41 heavy-duty configuration incorporates a hollow shaft input with an integrated, high-capacity predesigned sheave assembly supplied by the Factory, explicitly engineered for reliable horizontal mounting drives. Size 1420HF is rigidly secured directly to the motor shaft utilizing a heavy-duty collet and collet draw bolt assembly. Sizes 1480HF and 1584HF are robustly anchored via an axial retention fastener assembly. For these larger configurations, the hollow shaft output is bored to order and incorporates a modified transitional clearance profile fit with no setscrews. The driven shaft end must be precisely drilled and tapped for the retention fastener, and the package features a precision Factory-furnished spacer tailored to exact customer operational metrics.

AVAILABILITY BY COUPLING SIZE & TYPE

Type	Description	1420	1480	1584
HFN41	**Non Delay Fill**	X	X	X
HFD41	**Delay Fill**	X	X	X

HEAVY-DUTY LAYOUT & MOUNTING SCHEMATICS



Predesigned Standard Sheaves for Model 1420HF41:
• Part Number 0428352: 9.75" OD, 5V-5 Groove | • Part Number 0428231: 9.75" OD, 5V-8 Groove
• Part Number 0428353: 11.8" OD, 5V-5 Groove | • Part Number 0428162: 11.8" OD, 5V-10 Groove

TYPE HF41 PHYSICAL PARAMETERS & BORES

TABLE 1: GENERAL ENVELOPE & MOUNT DIMENSIONS

CPLG SIZE ★	Cplg Weight lb No Bore, W/O Sheave, W/O Fluid		Allowable Speed rpm	Dimensions — Inches					IA Fastener Size	QA (Max Shaft Eng)	SE ‡ (Min)	CPLG SIZE ★
	HFN	HFD		D	G	N	AA	BA				
1420HF	160	168	1800	2.35	6.74	4.71	18.70	19.25	—	—	4.38 ★	1420HF
1480HF	159	172	1800	...	8.07	2.99	21.65	19.80	7/8-9 UNC	8.25	5.62	1480HF
1584HF	276	295	1800	...	8.07	2.91	26.38	21.30	1-8 UNC	10.00	6.25	1584HF

TABLE 2: SHAFT CAPACITIES & ROTATIONAL INERTIA (WR^2) PROFILE

FLUID CPLG SIZE ★	Min Bore † Inches	Max Bore — Inches		WR^2 ($lb\text{-}in^2$) ♦ — Type HFN			WR^2 ($lb\text{-}in^2$) ♦ — Type HFD			FLUID CPLG SIZE ★
		Square Key	Rectangular Key ■	Input	Output	Fluid @ Max Fill	Input	Output	Fluid @ Max Fill	
1420HF	1.875 †	3.000	3.375	663	2,460	820	663	2,529	990	1420HF
1480HF	2.062	3.750	4.125	1,230	4,440	1,504	1,230	4,615	1,980	1480HF
1584HF	2.250	4.000	4.375	2,840	9,910	3,844	2,840	10,080	4,955	1584HF

★ **General Operational & Formatting Notes:** Fluid couplings are offered for horizontal mounting configurations as shown. Refer to the Factory or your local Authorized India Partner for other specialized mounting arrangements. Dimensions are for reference only and are subject to change without notice unless certified. For Model 1420HF specifically, please reference Page 31 of the main catalog for detailed values regarding dimensions K, L, and SE (Max).

† **Minimum Bore Configurations:** Minimum bore is the smallest bore to which the internal locking collet will be machined by the Factory. Standard collet definitions are indexed on Page 31. Please consult the Factory or local partner for unlisted non-standard collet profiles.

‡ **Preferred Shaft Engagement Capacity:** The preferred baseline engagement metric between the driving motor shaft and the internal hollow bore is engineered to equal twice the absolute diameter of the input driving shaft.

■ **High-Capacity Rectangular Keys:** To utilize these specific maximum recorded bores, a customer-furnished rectangular key must replace the standard square key supplied with the shaft. Always **CHECK COMPONENT KEY STRESSES** during operational configuration.

♦ **Mass Moment Intertial Constants:** Equivalent mass moment values shown apply strictly to the fluid coupling unit (without bore), and matching sheave assembly. Refer to Page 31 Engineering Data for fluid WR^2 multipliers for other than maximum fill parameters.

ENGINEERING DESK & COMMERCIAL QUOTES

For direct confirmation of specific sheave grove metrics, collet tolerance questions, or priority price inquiries, connect directly with the specialized engineering desk at **Jinharsh Industrial Solutions Private Limited**.